

Technical Document #602: Retrieval Augmented Generation - Comprehensive Overview

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Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Chunking splits documents into smaller segments for embedding.
- Re-ranking models score retrieved documents for relevance to the query.
- Vector databases store dense embeddings for semantic search.